Year 1 AS Statistics and Mechanics

Chapter 1

Data collection

Practice Booklet

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1.1 Populations and samples

Practice Book, printed page 1.

1. Explain what is meant by:

a. a population

b. a census

c. a sample.

Hint

A national census is taken every 10 years in the United Kingdom.

2. Explain what is meant by:

a. a sampling unit

b. a sampling frame.

Hint

Number plates form a sampling frame for all cars in the United Kingdom.

3.

a. State one advantage and two disadvantages of taking a census.

b. State two advantages and one disadvantage of taking a sample.

Hint

Researchers have to balance time, cost, convenience and reliability of results when choosing methods of data collection.

4. E A sports club with 713 members wants to investigate the types of exercise its members do outside of the club.

a. Suggest, with a reason, whether it is sensible to use a census to collect the data. (2 marks)

b. Suggest an alternative method that the sports club could use to collect the data and give one advantage of this method. (2 marks)

5. E A supermarket manager wants to test the softness of loaves of bread in the store bakery by squashing them.

a. Suggest a reason why the manager should not test all the loaves of bread. (1 mark)

The manager decides to take a sample of 4 loaves and finds that 2 of the loaves meet the required standard of softness. The manager concludes that only half of the loaves in the bakery meet the standard.

b. Give a reason why the manager's conclusion might be unreliable. (1 mark)

c. Suggest one way that the manager can improve the reliability of the conclusion. (1 mark)

6. E A local council wants to know what people think about the traffic flow in a particular town. The transport department decides to carry out a survey of the residents.

a. Suggest one reason why the transport department should not take a census. (1 mark)

The transport department decides to choose a sample of residents who will complete the survey.

b. Identify the sampling units. (1 mark)

The transport department decides to use the electoral register as a sampling frame.

c. Give one disadvantage of using this sampling frame. (1 mark)

7. E/P A manufacturer of light bulbs is testing the lifetime of the bulbs. The manufacturer switches the bulbs on and records the number of hours each bulb is lit before it blows.

a. Explain why the manufacturer should take a sample rather than a census. (1 mark)

The manufacturer tests a sample of three bulbs and obtains the following results:

762 hours 851 hours 801 hours

b. The company claims that the lifetime of its bulbs is 800 hours.

i. Use the sample data to comment on this claim.

ii. Suggest one way in which the manufacturer can improve its prediction about the lifetime of a bulb.

(3 marks)

1.2 Sampling

Practice Book, printed page 2.

1.

a. Explain what is meant by a simple random sample.

b. Explain how you would take a simple random sample of size 20 from a population of 100.

Hint

You can use your calculator to generate random numbers.

2.

a. Explain what is meant by a systematic sample.

b. Explain how you would take a systematic sample of size 20 from a population of 100.

Hint

Work out the proportion of the population that you are going to sample.

3.

a. Explain what is meant by a stratified sample.

b. Explain how you would take a stratified sample of size 20 from a population composed of 40 males and 60 females.

Hint

Work out the proportion of males and females in the population.

4. E/P A dance school teacher wants to take a sample of the school's members. Each member has a unique ID number.

a. State two advantages of using simple random sampling. (2 marks)

There are 75 boys and 125 girls at the school.

b. Using this information, state one disadvantage of using simple random sampling. (1 mark)

The teacher decides to use a stratified sampling method.

c. Explain how the teacher can take a stratified sample of size 40. (2 marks)

5. E/P A company wants to take a sample of its employees. There are 700 employees and each has a unique employee number.

a. Explain how the company could take a systematic sample of size 35 from the employees. (2 marks)

It is known that the management team all have employee numbers ending in 0.

b. Using this information, state one disadvantage of using a systematic sample. (2 marks)

6. E/P A school wants to carry out an investigation into the eating habits of its students. There are 1200 students in the school.

a. Explain how the school could take a simple random sample of size 50 from the students and state one disadvantage of using this sampling method. (2 marks)

b. Suggest a better sampling method and explain how the school should take this sample. (3 marks)

1.3 Non-random sampling

Practice Book, printed page 3.

1.

a. Explain what is meant by opportunity sampling.

Researchers want to investigate the habits of shoppers in a town's high street.

b. Explain how they might select an opportunity sample of size 20.

Hint

Opportunity sampling and quota sampling are examples of non-random sampling methods. Using non-random sampling methods can introduce bias.

2.

a. Explain what is meant by quota sampling.

Researchers want to investigate people's attitudes to advertising in the cinema.

b. Explain how they might select a quota sample of size 30.

Hint

Make sure your answer refers to the context of the question.

3. E/P A coach of a basketball team wants to take a sample from all the players on the team. The coach has a list of the players in order of height.

a. Explain how the coach can carry out an opportunity sample of size 5. (1 mark)

b. State one disadvantage of using an opportunity sample in this context. (1 mark)

c. Suggest an alternative sampling method that could be used and explain how the coach can take this sample. (2 marks)

4. E/P Researchers are collecting data on the number of hours of television watched by people in a local village. They ask the first four people they see coming out of the cinema on a Tuesday evening.

a. Describe the sampling method used and comment on the reliability of the data gathered. (2 marks)

b. Suggest two ways to improve the method used by the researchers. (2 marks)

5. E/P A researcher wants to investigate the types of car owned by people living on a housing estate. It is known that there are 150 male drivers and 200 female drivers on the particular estate.

a. Explain how the researcher can take a quota sample of size 70. (2 marks)

b. State one key difference between quota sampling and stratified sampling. (1 mark)

c. Give one advantage of using quota sampling over stratified sampling. (1 mark)

The researcher stands at the entrance to the estate. For each car entering the estate, the researcher notes the gender of the driver and the type of car.

d. Criticise this method of collecting the data. (2 marks)

1.4 Types of data

Practice Book, printed page 4.

1. State whether the following variables are qualitative or quantitative:

a. height of a house

b. hair colour

c. time taken to run 50 m.

Hint

Quantitative data can be recorded as a numerical observation. **Qualitative** data must be recorded as words or categories.

2. State whether the following variables are discrete or continuous:

a. length of a worm

b. number of slugs

c. height of an ant.

Hint

Variables that can take any value in a given range are **continuous**. Variables that can only take specific values are **discrete**.

3. E The table shows the masses of children on a summer camp.

Mass, x (kg)	**Frequency**
$35 \leq x < 40$	14
$40 \leq x < 45$	18
$45 \leq x < 50$	12
$50 \leq x < 55$	7

a. Explain why the data in the table is continuous. (1 mark)

b. For the third group, write down:

i. the class boundaries

ii. the class width

iii. the midpoint.

(3 marks)

4. E/P The table shows the numbers of caterpillars found on each tree in a forest.

Number of caterpillars	**Number of trees**
7	78
8	45
9	63
10	18

a. State whether this data is discrete or continuous. (1 mark)

A researcher wants to see if there is a relationship between the number of caterpillars on each tree and the lengths of the caterpillars.

b. Explain how the researcher can take a stratified sample of size 20 from all of the trees in the forest. (2 marks)

The researcher measures the lengths of the caterpillars on each of the sampled trees.

c. Describe the type of data the researcher is collecting. (1 mark)

1.5 The large data set

Practice Book, printed page 5.

1. From the five UK weather stations featured in the large data set, write down the station that is furthest:

a. north

b. south

c. west.

Hint

You need to know the approximate locations of all the weather stations in the large data set.

2. Explain what is meant by:

a. daily total rainfall

b. daily maximum gust.

Hint

You need to know the units for all of the data set variables.

3. E/P Delilah is interested in how the daily total sunshine changes the further north you go. She chooses five days, at random, from the large data set for Heathrow and Hurn in 2015.

a. Criticise:

i. her sample size

ii. her choices of location.

(2 marks)

b. Explain how Delilah can take a simple random sample of size 15 from the large data set for Heathrow in 2015. (2 marks)

During her sampling, Delilah selects the same date twice.

e. Explain why this is wrong and describe how she could correctly generate her sample. (2 marks)

4. E/P From the large data set, Jerome calculates the average mean daily temperature for Leuchars and Camborne in 2015 using a random sample of 5 days from each location.

a. Give a geographical reason why the average mean daily temperature for Leuchars might be lower than for Camborne. (1 mark)

b. Criticise Jerome's sampling method and suggest an improvement. (2 marks)

5. E/P Amara is investigating daily total rainfall in Leeming in 2015.

a. Explain how she can take a systematic sample of size 20 from the data. (2 marks)

She finds that on several days, the reading is recorded as 'tr'. She discards these readings and chooses additional readings for her sample.

b. State the meaning of 'tr', and evaluate Amara's method. (2 marks)

c. Suggest how Amara should deal with 'tr' readings in her investigation. (1 mark)

6. E/P Jagdeep records the daily mean windspeed for a randomly chosen day in June 1987 in Hurn as 31 knots. He records the daily maximum gust on the same day as 25 knots.

a. Explain why Jagdeep must have recorded at least one of the values incorrectly. (1 mark)

The correct value for the daily mean windspeed is in fact 11 knots.

b. Convert this reading using the Beaufort scale, giving your answer as both a number and a description. (2 marks)

7. E/P A researcher is investigating average weather conditions in July 2015 in Beijing, Perth and Hurn. The researcher records the following data:

Location	A	B	C
Average temperature (°C)	16.8	26.8	13.3
Average windspeed (kn)	8.3	4.0	7.4

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Using your knowledge of the large data set, suggest the name of the weather station that is recorded as:

a. location A

b. location B

c. location C.

(3 marks)

Problem solving: Set A

Bronze

Joanna is investigating the daily total rainfall in Perth in 2015. She uses the large data set to select a sample from all 184 days.

a. Describe how Joanna could take a random sample of 5 days. (2 marks)

b. Give one advantage and one disadvantage of using a random sample. (2 marks)

c. Suggest one improvement that Joanna can make to improve the reliability of the data she collects. (2 marks)

Silver

Francine is investigating the daily mean windspeed for Beijing for August, September and October 2015. She uses the large data set to select a sample of size 20 by generating a random starting number between 1 and 5 and choosing every fifth value thereafter.

a. Describe the sampling method that Francine has used. (1 mark)

b. Using your knowledge of the large data set, explain why Francine's method will not generate a sample of size 20. (2 marks)

Gold

Ali is investigating the daily mean wind direction in May, June and July in Jacksonville in 2015. He uses the large data set to select a sample from all 92 days.

a. Describe how Ali can select a sample of size 15 stratified by month. (2 marks)

b. Give one advantage of using a stratified sample in this context. (1 mark)

c. Describe the two ways that Ali's data is recorded in the large data set. (2 marks)

Problem solving: Set B

Bronze

Simon is researching the eating habits of the students in his school year. He asks the first five people he sees on Monday morning.

a. Write down the sampling technique that Simon has used. (1 mark)

b. State one advantage of this technique. (1 mark)

c. Suggest two improvements that Simon can make to the sampling technique. (2 marks)

Silver

Corina is researching the musical tastes of the students in her school. She is considering two types of sampling technique: opportunity sampling and systematic sampling.

a. State which of these two sampling methods is an example of random sampling. (1 mark)

b. Give one advantage and one disadvantage of opportunity sampling. (2 marks)

c. Give one advantage and one disadvantage of systematic sampling. (2 marks)

d. Advise Corina on which of the two methods she should use, and give a reason for your choice. (2 marks)

Gold

Ishant is researching the shopping habits of people in his local town. He is going to carry out his research using quota sampling and decides to stand in the middle of the market square in his town at 2 pm on Saturday.

a. Suggest a reason why his chosen time and location is a good one. (2 marks)

Ishant can legally only ask adults.

b. Suggest suitable groups for Ishant's sample. You should specify at least four groups. (2 marks)

c. Suggest a method by which Ishant could determine a suitable quota for each group. (2 marks)

Now try this → Exam question bank Q1, Q3, Q7, Q17

Selected exam question bank questions

Question 1

Alice wants to survey the students in her year group to find out their favourite type of animal. There are 75 boys and 85 girls in her year.

Explain briefly how Alice could take a random sample of 40 pupils using:

a. systematic sampling (2 marks)

b. stratified sampling. (2 marks)

Question 3

Eleri is investigating the daily total sunshine in Hurn during June 1987. She uses data from the first five days of the month.

a. Describe the sampling method used by Eleri. (1 mark)

b. Suggest two alternative sampling methods and give one advantage and one disadvantage of each method in this context. (2 marks)

c. State, with a reason, whether the data that Eleri has collected is discrete or continuous. (1 mark)

Question 7

A manufacturer of bicycle inner tubes wants to test the pressure at which the tubes burst.

a. Explain why it would not be sensible for the manufacturer to use a census. (1 mark)

The manager decides to take a random sample of 5 inner tubes and measure the pressures at which they burst. The data, in kilopascals (kPa), is recorded below:

525 567 610 660 680

The manager says that the data supports packaging the inner tubes with a label which states: 'Inner tubes can withstand a pressure of at least 600 kPa.'

b. Use the data above to suggest a reason why this labelling is not suitable. (2 marks)

c. Suggest one way in which the manager can improve the reliability of the sampling. (1 mark)

Question 17

a. Explain what is meant by a sampling unit. (1 mark)

b. Explain what is meant by a sampling frame. (1 mark)

Manvitha is investigating the daily mean windspeed in Perth during 2015. She uses data from all 184 days in the large data set and decides to take a simple random sample of size 20.

c. Describe how Manvitha could take her sample. (2 marks)

d. Give two advantages of using a simple random sample. (2 marks)

Answers

1.1 Populations and samples

1.

a. The whole set of items that are of interest.

b. A census observes or measures every member of a population.

c. A selection of observations taken from the population.

2.

a. An individual unit of a population.

b. A (numbered) list of the sampling units.

3.

a. Advantage: it should give a completely accurate result.
Disadvantages: time consuming and expensive; cannot be used when the testing process destroys the item; hard to process large data sets.

b. Advantages: less time consuming and expensive; less data to process. Disadvantage: the data may not be representative of the population.

4.

a. Not sensible: too time consuming.
Sensible: will use all of the data and give an accurate result.

b. Take a sample. Advantages: quicker and cheaper; less data to process afterwards.

5.

a. The loaves will be damaged by squashing them so there will be none left to sell.

b. A very small sample size, so may not be reflective of the overall proportion.

c. Taking a larger sample will improve the reliability.

6.

a. Time consuming/costly; hard to process all of the data.

b. The individual residents of the town.

c. The electoral register of voters does not include people who are ineligible to vote, such as those under the age of 18, and may also be incomplete since there may be residents who have not registered to vote.

7.

a. A census will destroy all the bulbs.

b.

i. While the average is greater than 800, one of the bulbs tested lasted less than 800 hours; so the claim is justified on average, but not on the basis that it refers to all of the bulbs.

ii. Increase the number of bulbs that are sampled.

1.2 Sampling

1.

a. A simple random sample of size n is one where every sample of size n has an equal chance of being selected.

b. Put the population into an ordered list (sampling frame) and generate 20 random numbers using the random number function on a calculator.

2.

a. The required items are chosen at regular intervals from an ordered list. The first item is chosen at random.

b. Choose the first item randomly (such as by using a random number generator) and then select every 5th item from the population.

3.

a. A sample which is proportional to the number of items in each strata/group.

b. Calculate the proportion of each group required in the sample (8 and 12) and then select these at random from an ordered list.

4.

a. Easy and cheap to carry out; each pupil has a known and equal chance of being selected.

b. The simple random sample may not reflect the proportions of boys and girls at the school.

c. The teacher should divide the sample size into the same proportion as the population and then randomly select the (15) boys and (25) girls.

5.

a. Generate the number of the first employee to be chosen randomly and then select every 20th employee from the ordered list of employee numbers.

b. The sampling frame is not random so the sampling method can introduce bias, for example if the first person chosen has a number ending in 0, all of the employees chosen will have a number ending in 0 and the entire sample will consist of managers.

6.

a. Number the students, perhaps by using their admission number, and then randomly generate 50 numbers. Disadvantages include the fact that the population is (fairly) large and that it might not be representative of the different age/year groups of pupils.

b. Stratified sampling; divide the sample into the same proportion as the proportion of students in each year (possibly split male/female too) and choose the required number of students randomly from each individual group.

1.3 Non-random sampling

1.

a. Opportunity sampling consists of taking the first n people who are available at the time of the research.

b. Stand in the street and ask the first 20 people who pass by and are prepared to answer questions.

2.

a. The sample is chosen so that it reflects the characteristics of the whole population.

b. Allocate quotas to each film showing at a large cinema. Select people who buy tickets for each film in order until the quota is fulfilled.

3.

a. Take the first 5 players from the ordered list.

b. These players will be the five shortest (or tallest) players and are unlikely to be representative of the team as a whole.

c. Take a simple random sample by generating five random numbers and selecting these players from the (numbered) ordered list.

4.

a. Opportunity sampling; the data is unlikely to be representative since the researchers ask a very small sample of people in a very specific location.

b. Take a larger sample of people; vary the time, day and location of the research.

5.

a. Divide the sample into the same proportion as the population to create each quota (30 males and 40 females), assign each person the researcher meets to a quota, then continue to ask/observe people until each quota is full.

b. In a stratified sample, the people selected in each group will be generated randomly; in a quota sample they will be the first n people the researcher meets.

c. A stratified sample needs a sampling frame, which may not be available to the researcher.

d. The researcher does not know that the driver owns the car that is being driven and does not know that the driver actually lives on the estate.

1.4 Types of data

1.

a. Quantitative

b. Qualitative

c. Quantitative

2.

a. Continuous

b. Discrete

c. Continuous

3.

a. It is measured and can take any value.

b.

i. 45 and 50

ii. 5

iii. 47.5

4.

a. Discrete

b. Divide the sample in the same proportion as the different strata (number of caterpillars) and choose from each stratum at random. Proportions are 8, 4, 6, 2.

c. Continuous (quantitative) data

1.5 The large data set

1.

a. Leuchars

b. Camborne

c. Camborne

2.

a. Amount of precipitation including snow and hail that falls during a 24-hour period; measured in mm.

b. Highest instantaneous windspeed recorded on a particular day; measured in knots.

3.

a.

i. It is too small.

ii. It uses two locations that are almost the same latitude as each other.

b. She can assign each date a number and then use a random number generator to generate the 15 dates in the sample.

c. The sample should contain 15 different dates: data values cannot be reused. She should therefore reject the repeated value in the random number generation and generate a 16th random number.

4.

a. Leuchars is further north.

b. The sample size is very small; Jerome should increase the sample size to give a more representative sample.

5.

a. Generate a random starting number then count on every 9 dates (since $\frac{184}{20} \approx 9$).

b. The notation 'tr' means trace (less than 0.05 mm). Amara's method is not suitable as she has discarded valid data values.

c. Any data value given as 'tr' should be recorded as 0 for the purpose of carrying out any calculations.

6.

a. Daily maximum gust must be greater than or equal to the daily mean windspeed.

b. 4; Moderate

7.

a. Hurn

b. Beijing

c. Perth

Problem solving: Set A

Bronze

a. Generate five random numbers and select the dates from a numbered (ordered) list.

b. Advantage: it is quick and easy to carry out.
Disadvantage: it may not be representative of the whole data set.

c. Take a larger sample.

Silver

a. Systematic sampling.

b. There are only 92 days in these three months so her sample will be of size 18 or 19 (depending on the starting number).

Gold

a. Select five days at random from each month using a random number generator.

b. The data is bound to be taken across all three months, ensuring proportional representation of each month.

c. As a bearing; as a cardinal (compass) direction.

Problem solving: Set B

Bronze

a. Opportunity sampling.

b. Quick and easy to carry out.

c. Increase the number of people he asks; vary the time of day he asks people.

Silver

a. Systematic sampling.

b. Advantage: quick and easy to carry out.
Disadvantage: not necessarily representative (limited time/location).

c. Advantage: is likely to be representative of the students as a whole. Disadvantage: a sampling frame is required.

d. Systematic sampling: a sampling frame is likely to be available – a register of students, for example.

Gold

a. It is in the middle of the day on a weekend, so more people are likely to be out shopping; the chosen location is central.

b. For example, male shoppers under 40, female shoppers under 40, male shoppers 40 and over, female shoppers 40 and over.

c. For example, use the electoral register to determine the size of each group as a proportion of the whole population, then assign the quotas as the same proportion of the whole sample.

Selected exam question bank answers

Question 1

a. Get a numbered (ordered) list of the pupils, choose the first one at random and then pick every fourth pupil.

b. Choose at random from the lists of boys and girls in proportion to the total of each gender (19 boys, 21 girls).

Question 3

a. Opportunity sampling.

b. Simple random sampling. Advantage: quick and easy; disadvantage: might not fairly represent the whole month.
Systematic sampling. Advantage: representative across the whole month; disadvantage: might not get a sample of size 5 since data could be missing.

c. Continuous, as it can take any value within the range.

Question 7

a. The testing is to destruction so all of the inner tubes would be damaged.

b. While the average number of kPa is greater than 600, two-fifths of the sampled inner tubes burst below this pressure.

c. Sample more inner tubes.

Question 17

a. An individual unit of a population.

b. A numbered/ordered list of the sampling units in a given population.

c. Generate 20 random numbers and then use those numbers to read off the data for the corresponding dates in the (ordered) list of dates.

d. Free of bias; easy/cheap/quick to carry out; each sampling unit has a known and equal chance of being chosen.